BioBead Greenhouse Study: Analytical Methods

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Introduction

The literature on the effects of biofertilizer on different soil geochemical properties was reviewed to understand best data analysis practices.

Demir (2020) tested the effects of a biofertilizer on soil physiochemical properties. The data were analyzed with a one-way analysis of variance (ANOVA) followed by a Pearson's correlation coefficient assessment. Harroun and colleages (2023) also tested a biofertilizer on soil physiochemical properties (with a heavy metal remediation focus). They also used ANOVA (with means compared with Fisher LSD test) followed by a Pearson's correlation coefficient assessment. Rajper et al (2016) tested a soil inoculant on soil microbes and enzymes. They used a repeated-measure one-way ANOVA to analyze these data at the different timepoints, and a Pearson correlation coefficient analysis to explore the correlations between soil microbes and enzymes activity. Then they used a Principal component analysis (PCA) for phospholipid fatty acid analysis to evaluate the microbial biomass and community structure. Shen et al (2015) tested the effects of a biofertilizer on microbial and chemical properties. This paper had more of a microbiological focus. They used a one-way ANOVA to evaluate the microbial communities associated with different fertilizer treatments. They used Pearson and Spearman correlations for the top bacterial genera and soil properties. They also did a PCA for different fertilizer regimens, as well as a redundancy analysis (RDA) on the relative abundance of bacterial phyla and different soil chemical properties. Wu et al (2004) tested a biofertilizer and evaluated the effects on soil physiochemical properties. The data were analyzed with a one-way analysis of variance (ANOVA) followed by a Pearson's correlation coefficient assessment. Finally, Zhao et al (2019) tested a soil inoculant on soil microbes, nutrients, and chili plant growth. They used a Student's t test to find significant differences in soil properties, nutrient accumulation, bacterial alpha diversity, and relative abundances of different taxa. They also used non-metric multidimensional scaling (NMDS) ordination plots to evaluate the bacterial community structure. Redundancy analysis (RDA) was also conducted to analyze the relationships between microbial community structure and different soil environmental factors. All of these studies were from a greenhouse setting.

In summary, these research groups commonly used **single factor ANOVAs**. Post-hoc tests like Tukey HSD were not mentioned. Some did not go beyond univariate analyses. **Pearson's correlation coefficient** was used in some studies to determine which response variables related to one another. For groups that ventured into multivariate analyses, P**CAs and RDAs** were the techniques that were chosen.

Analytical approaches like PERMANOVA and linear mixed effects models are used in other fertilizer/agricultural soil studies. Soonvald et al, (2019) used 9999 permutations in their analysis. Utilizing these analyses in a greenhouse study of biofertilizers would enhance the literature on microbial inoculants and soil effects and be a powerful contribution.

Data matrix

The data are compiled into a single spreadsheet (**2025greenhousemaster.csv**), which can be found in the GitHub repository. It was compiled from the following raw data.

Leachates_2025.csv - Ammonium, nitrate, and phosphate leachate concentrations recorded throughout the study.

PRS_GH_2025.csv - Soil anion and cation exchange data from PRS probes.

WheatMoistureData_2025.csv - Soil moisture data recorded throughout the study.

N2O_flux.data.csv - Nitrous oxide (N2O) flux was also calculated. The data are limited to 3 replicates per treatment (not 7), so these data will remain in its own file rather than grafted to the master spreadsheet.

Wheat yield data has been added to the master spreadsheet, and includes 2 columns on grain yield (per sample unit) and an associated Harvest Index (HI), which is a ratio of grain yield over the rest of the above-ground biomass. It represents an economically-relevant measure of wheat crop productivity.

Exploratory one-way ANOVAs have already been performed on leachate, soil moisture, anion/cation supply rate, wheat yield, and N2O flux.

Additional response variables

Table 1: Table 1: Additional response variables. Original response variables can be found in the project overview.

Code	Full Name	Units	Range
GrainWeight_g	Grain weight	g	0.6-2.7
Harvest_Index	Harvest index	n/a	0-0.6
N2O-N_flux	Nitrous oxide N flux	ug N2O-N/sqmt/hr	-38-3557

Research Questions

1. Does the hydrogel carrier in BioBead explain variation in the soil geochemical profile?
2. How do treatment effects on soil properties develop over time?
3. How do microbe type and strigolactone priming interact to affect soil nutrient dynamics?

Future analyses will explore the N2O-N fluxes, which are significant in this data set. Additionally, an analysis focused on wheat yield (specifically, which factors maximize it) is warranted. Metagenomic data is incoming as well, allowing connections with these data and microbial community composition.

Statistical approaches

Hydrogel influences on soil geochemical profile

BioBead contains bacteria and fungi in a biodegradable hydrogel carrier. The hydrogel holds water, breaks down into C-rich oligosaccharides, and keeps the microbes at the root where they are most important. I hypothesize that the hydrogel carrier is important for the technology.

To determine this, I can use a **PERMANOVA** on the multivariate soil profile at Week 6. This timepoint is selected because there were more significant effects detected at this timepoint in the one-way ANOVAs, and the soil anion/cation exchange data is from that sampling period. Week 6 is also when the microbes are established, so it's the most ecologically relevant timepoint. Euclidean distances would be used to create the distance matrix. The data would definitely need to be standardized (the scales are wildly different).

I would then follow this up with a PERMANOVA focusing on the hydrogel treatments in the study to test the hypothesis that the hydrogel carrier is associated with variation in soil physiochemical properties in this study. I would use 9999 permutations for publication-quality results on both of these.

Finally, I could bring in **RDA** to see what drives differentiation of these different responses.

Temporal analysis of soil properties

A **repeated measures ANOVA** can be used to analyze leachate, N2O-N flux, and soil moisture data over time. This involves similar assumptions to ANOVA: independent observations, normal residuals, and sphericity instead of equal variances. Residual normality and sphericity will be checked with QQ plots and Mauchly's test respectively.

An alternative approach could be linear mixed effects models. But since this design is balanced with even timepoints, a repeated measures ANOVA will suffice.

Strigolactone and microbial interactions

It would be interesting to investigate the effects of strigolactone priming on key nutrients in the data set. The structure would be similar to that used for the hydrogel factor analysis: a PERMANOVA focusing on strigolactone (GR24) presence or absence in a set of treatments.

References

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